

TimedExec Test Cases

Command name: timedexec

Compile command: gcc timedexec.c -o timedexec

Test Case 1: Normal Execution (No limits exceeded)

Command:

`./timedexec -t 5 ls -l`

Expected Output:

(Standard output of the ls -l command showing directory contents)

[timedexec] Command exited normally with status 0

Purpose:

This test checks that the program correctly runs a standard command without interfering when limits are not hit.

Test Case 2: CPU Time Limit Exceeded

Command:

`./timedexec -t 2 bash -c "while true; do echo 'Running...'; done"`

Expected Output:

(Prints "Running..." rapidly for exactly 2 seconds)

[timedexec] KILLED: Exceeded CPU time limit of 2 seconds.

Purpose:

This test checks if the setrlimit() successfully triggers a SIGXCPU signal from the kernel when the process hogs the CPU, and that the parent process correctly interprets this signal.

Test Case 3: Missing Command Argument

Command: `./timedexec -t 5 -m 100000`

Expected Output:

Error: No command specified to execute.

Usage: `./timedexec [-t cpu_time_seconds] [-m memory_limit_bytes] command [args...]`

Purpose:

This test checks `getopt()` argument handling to ensure the program doesn't crash if the user forgets to type the actual program they want to run.